

Write SystemVerilog HDL to leverage the internal high speed oscillator in the Lattice UP5K FPGA and blink an LED at 2 Hz. Take full advantage of SystemVerilog syntax, for example, using logic instead of wire and reg.

For the UPduino v3.1 / iCE40 UP5K, you can use the FPGA's internal `SB_HFOSC` oscillator and a counter to divide it down.

With `CLKHF_DIV = 2'b01`, the nominal oscillator frequency is **24 MHz**. To blink at **2 Hz**, if by "blink" you mean the LED completes **2 full ON/OFF cycles per second**, the LED needs to toggle at **4 Hz**:

$$N = \frac{24\,000\,000}{4} = 6\,000\,000$$

```

module blink_2hz (
    output logic led
);

</> systemverilog

// Internal high-frequency oscillator
// 48 MHz / 2 = 24 MHz
SB_HFOSC #(
    .CLKHF_DIV("0b01")
) hfosc (
    .CLKHFPU(1'b1),
    .CLKHFEN(1'b1),
    .CLKHF(clk)
);

always_ff @(posedge clk) begin
    if (counter == 23'd5_999_999) begin
        counter <= '0;
        led    <= ~led;
    end
    else begin
        counter <= counter + 1'b1;
    end
end

```

```

        end
    end

endmodule

```

## Why 6,000,000 counts?

The internal clock is:

$$f_{clk} = 24 \text{ MHz}$$

For a 2 Hz blink:

```

ON    = 0.25 s
OFF   = 0.25 s

```



```

Full period = 0.50 s
Frequency    = 2 Hz

```

Since the LED changes state every 0.25 s :

$$24,000,000(0.25) = 6,000,000$$

So the counter counts from 0 through 5,999,999 , then toggles led .

One small improvement for a lab assignment would be to parameterize the clock and blink frequency instead of hard-coding 5\_999\_999 .



| synthesis (1 errors, 2 info) |       |                                                                               |
|------------------------------|-------|-------------------------------------------------------------------------------|
| 35001797                     | Info  | Use DCC insertion disabled by default.                                        |
| 35901018                     | Info  | ../source/impl_1/top.sv(1): compiling module top. VERI-1018                   |
| 35901063                     | Error | ../source/impl_1/top.sv(16): instantiating unknown module SB_HFOSC. VERI-1063 |

failed to synthesize

Yep — the error tells us exactly what's wrong:

```
instantiating unknown module SB_HFOSC
```

Your synthesis environment for the UP5K expects the Lattice primitive HSOSC , not SB\_HFOSC . SB\_HFOSC is the name commonly used with the open-source iCE40/Yosys flow,

while your current Lattice synthesis flow recognizes HSOSC .

Use this version:

```
</> systemverilog
module top (
    output logic led
);

    logic clk;
    logic [22:0] counter = '0;

    // UP5K internal high-speed oscillator
    // 48 MHz / 2 = 24 MHz
    HSOSC #(
        .CLKHF_DIV(2'b01)
    ) osc (
        .CLKHFPU(1'b1),
        .CLKHFEN(1'b1),
        .CLKHF(clk)
    );

    always_ff @(posedge clk) begin
        if (counter == 23'd5_999_999) begin
            counter <= '0;
            led <= ~led;
        end
        else begin
            counter <= counter + 1'b1;
        end
    end

endmodule
```

The important change is simply:

```
</> systemverilog
SB_HFOSC
```

→

```
</> systemverilog
HSOSC
```

Your `CLKHF_DIV = 2'b01` gives a **24 MHz clock**, so counting 6,000,000 clock cycles before each LED toggle gives:

$$\frac{24 \text{ MHz}}{6\,000\,000} = 4 \text{ toggles/sec}$$

and because one blink cycle requires two toggles:

2 Hz blink

Also, based on the compiler message, your module is currently named `top`, so keeping `module top` as above should fit your project directly.



ChatGPT can make mistakes. Check important info.